Level 3 costs more than level 2 on 45 of 217 circuits and PSEL4 removes that risk at 15.7 % less HQC

pytket-quantinuum's optimisation level 3 saves 13.9 % HQC over the default level 2 on our 217-circuit benchmark but costs more on 45 of them (372.0 HQC lost, worst 12.95 % on gf2_pow_4_mult), and a user cannot tell in advance which. PSEL4 removes that risk. It compiles each circuit four ways (level 2, level 3, ZX before each), checks each result and keeps the cheapest. It is costlier than level 2 on none of the 217, 15.7 % below level 2, 2.08 % (1,051.6 HQC) below always level 3 and 1.35 % (679.6 HQC) below the per-circuit cheaper of level 2 and level 3. A team that spends the Standard Plan's 10,000 HQC a month on this mix frees 1,573.3 HQC, worth USD 19,666 a month (workload assumption). [O]

Problem statement

A job on Quantinuum H2 is billed in HQC = 5 + N_shots (N_1q + 10 N_2q + 5 N_SPAM) / 5000. Level 3 and ZX rewriting can each lower the HQC of a circuit or raise it, so a cost-bound user needs the cheapest correct compile of each circuit without trying every option by hand. [O]

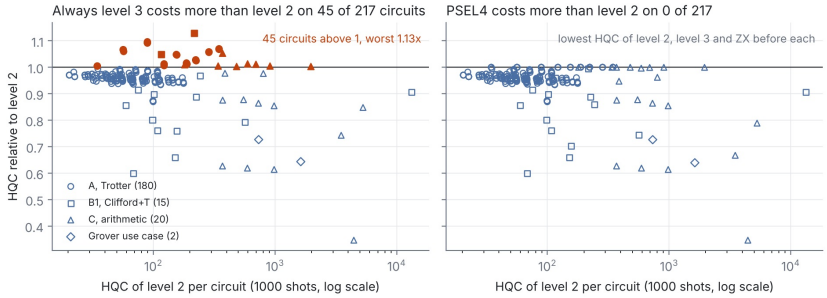
Approach

PSEL4 compiles each circuit four ways: level 2, level 3 and PyZX teleport_reduce before each. Each candidate is checked against its input as a full unitary up to 8 qubits and on 64 random computational basis inputs from 9 to 20 qubits. Above 20 qubits family C relies on PyZX miter proofs (result 4). PSEL4 drops a candidate that fails and keeps the lowest HQC.

Key results

1. Cost: level 3 adoption made safe. Total HQC at 1000 shots per circuit.

Policy, 217 circuits	Total HQC	Saved vs level 2	Circuits costlier than level 2	Largest per-circuit loss
Always level 2 (default)	58,837.2	0	0	none
Always level 3	50,632.2	13.95 %	45	12.95 % (gf2_pow_4_mult)
PSEL4	49,580.6	15.73 %	0	none



HQC relative to level 2 per circuit: always level 3 (left, the 45 regressions in orange) and PSEL4 (right, none above 1).

2. Fidelity on the H2-2E emulator. The HQC-cheapest compile is not the most accurate. Five pre-registered comparisons of the PSEL4 pick against level 2 on non-zero inputs (400 shots per program, 1000 on vbe_adder_7). The pick was significantly less accurate on 2 (cdkm_adder_10, level 3, 0.8267 against 0.8692, z = -2.90, and hwb6, ZX then level 3, 0.8883 against 0.9200, z = -2.64, Holm p = 0.0168 over the two circuits of its run), not significantly different on 2 (modadd_2n_10 z = -0.22, vbe_adder_3 z = +0.51) and not significantly better on 1 (vbe_adder_7 z = +1.30). On cdkm_adder_10 the less accurate pick has fewer two-qubit gates (118 against 141) and lower two-qubit depth (65 against 122, post hoc), so neither HQC nor gate counts order fidelity. The HQC saving still exceeds the accuracy loss: HQC per correct answer is lower for the pick in 5 of 5 pairs (8 of 8 including three earlier pairs of the two-candidate selector), pooled 18.1 % lower (R = 0.8192, bootstrap 95 % interval 0.8132 to 0.8254). PSEL4 serves cost-bound users. Accuracy-bound users need a fidelity-aware choice, which is the finals deliverable: a learned selector trained on this ledger and these emulator labels. [O]

3. Mechanism. HQC charges no Rz gate and bills every PhasedX at the same rate whatever its angle, so the T-count reduction ZX is best at has no HQC value on H2. ZX can pay mainly through two-qubit gates, which level 3 already resynthesises by GreedyPauliSimp, though phase teleportation still removes non-Clifford rotations that level 3 keeps (656 against 746 on 14 B1 circuits). On top of the cheaper pytket level the ZX candidates add 0.26 % on the 17 proven family C circuits, 3.21 % on all 20, 0.49 % on B1 and nothing on family A. The gain falls with T-density (family C Spearman rho -0.64).

4. Correctness. All 83 PSEL4 picks with 9 to 20 qubits pass the 64-basis-input check, as do 187 of 187 required candidates and 142 of 142 others, and negative controls fail as they should. A PyZX miter proves the PSEL4 output equal to its input on 17 of 20 family C circuits. MQT QCEC 3.10.2 proved none of the other three and its ZX and alternating checkers returned no counterexample on any pair. A conservative variant keeps the cheapest verified candidate, else the default level 2 output flagged "default, unverified" (only hrs_mult_5), saves 12.25 % against level 2 and is costlier than level 2 on 0 circuits.

Relevance to quantum computing

Adders sit in the modular arithmetic of Shor factoring and in arithmetic for derivative pricing (Chakrabarti et al., arXiv:2012.03819). PSEL4 lowers the cost of these subroutines on H2 today with no change to the user's code. [O]

Potential impact

The full ledger saving of 9,256.6 HQC is worth USD 95,289 to 115,707 at the Azure plan averages. [O] Drop-in use: `python3 pipelines/pass_select.py --psel4 <circuit.qasm>`, at 5.08 times the compile time of level 2. Submission 1 provides the ledger (217 circuits by 4 candidates, checked) and the emulator labels (5 pre-registered PSEL4-pick pairs, 48 jobs). A first classifier on input-circuit features does not beat a constant choice (0.433 against 0.500), so the model needs compiled-candidate features and emulator labels. [O]

Limits. The PSEL4 pick was significantly less accurate on 2 of 5 pre-registered pairs. Per-answer intervals pool inputs and ignore between-run variation. 3 of 8 per-answer pairs use P2, the earlier selector's pick. No user interview. Emulator only (48 H2-2E jobs, 9971.8 HQC). HQC is the formula estimate. Three PSEL4 and two level 3 family C outputs are unproven. On draper_adder_28 the PSEL4 pick starts from angles PyZX rounded by up to 1.50e-06 rad. The ledger run checked level 3 only up to 12 qubits. The mechanism analysis lacks 6 circuits.

Team Learned ZX Rewriting: C.-F. Kam, A. Mahasinghe, D. De Silva, N. Karunarathne, F. Cadet, M. Bessafi, K. De Silva. Quantinuum Singapore Grand Challenge, submission 1. Report `report.pdf`, appendix `report_full.pdf`, code `code.zip`, evidence `EVIDENCE.md`.

Untagged statements are facts checkable in the repository. [O] marks a judgement.